PYTHON DATA SCIENCE · BEFORE YOU BEGIN

# Set Up Your Data Science Workshop

About an hour of setup to do **before the first session**: Python 3.12 or newer, a virtual environment with the course packages, a notebook tool, and the Pinecrest Outfitters data. These are the same steps as Lesson 0.1 on the course site (rays-python-datascience.netlify.app), so if you get stuck, that page shows every step with its expected output. Session 1 then starts with data instead of installers.

⏱️ ~60 min, mostly downloads | Assumes Intro + Intermediate Python | Windows · Mac · Linux

> **PLEASE DO THIS AT HOME, NOT IN CLASS**
>
> The install downloads a few hundred megabytes. Venue Wi-Fi is rarely up to twenty people doing that at once. If a step fails, keep going with the others and bring the **exact error message** (a screenshot is fine) to Session 1: the first half hour is set aside for exactly this.

## WHAT YOU NEED

- **A laptop** (Windows, macOS or Linux) and its **charger**, with **permission to install software** and about **2 GB of free disk space**. Chromebooks, tablets and locked work laptops cannot run the course comfortably: talk to your instructor first.
- **Python 3.12 or newer**, a **terminal** (PowerShell on Windows, Terminal on macOS), and **JupyterLab** (installed with the course packages) or **VS Code**.
- **Google Chrome** (free). Selenium drives it in Session 17, and Plotly uses it to save charts as images in Session 19.
- **Two downloads** from the course site: `requirements.txt` and `pinecrest-data.zip`, plus `pinecrest-practice-site.zip` for the web sessions.
- **A learning journal**: a plain text file called `journal.md` or a paper notebook. Every lesson ends with a journal prompt.

## AM I READY FOR THIS COURSE?

This course follows **Introduction to Programming with Python** and **Intermediate Python** (both free at rays-home.netlify.app). No statistics or data experience is needed. You are ready if you can, without looking things up:

- ☐ Use variables, `if` / `elif` / `else`, and `for` and `while` loops
- ☐ Slice a list (`sales[2:5]`, `sales[-1]`) and say why `b = a` does not copy a list

- ☐ Build and loop over a dictionary with `.items()`, and count things with one
- ☐ Write a function with parameters, a default value and a `return`
- ☐ Read a list comprehension such as `[p * 2 for p in prices if p > 10]`
- ☐ Read a text or CSV file with `with open(..., encoding="utf-8")`, and use `import`
- ☐ Read a traceback and catch an error with `try` / `except`; write a small class with `__init__`

Two or more unchecked? Review those topics in the Intro and Intermediate courses before Session 1. Session 1 includes a short, ungraded diagnostic on these same skills, so your instructor knows where to help.

## SET UP IN 6 STEPS

### 1 Python 3.12 or newer

~5 min

Open a terminal and check what you have. A higher number than 3.12 is fine.

| SYSTEM | CHECK | MISSING OR OLDER THAN 3.12? |
|---|---|---|
| Windows | `py --version` | Install from **python.org/downloads**. In the classic installer, tick **"Add python.exe to PATH"** on the first screen. |
| macOS | `python3 --version` | The macOS installer from python.org, or with Homebrew: `brew install python@3.12` |
| Linux (Debian, Ubuntu) | `python3 --version` | `sudo apt install python3 python3-venv python3-pip`. Ubuntu 24.04 and newer give you 3.12. On an older release that still says 3.10 or 3.11, install `python3.12` and `python3.12-venv` (for example from the deadsnakes PPA) and use `python3.12` wherever Step 2 says `python3`. |

## 2 A project folder and a virtual environment

~5 min

Everything for the course lives in one folder, `pinecrest-analysis`, with its own environment inside it. Run these one line at a time:

| | WINDOWS (POWERSHELL) | MACOS / LINUX |
|---|---|---|
| Folder | `mkdir pinecrest-analysis` then `cd pinecrest-analysis` | |
| Create | `py -m venv .venv` | `python3 -m venv .venv` |
| Activate | `.venv\Scripts\Activate.ps1` | `source .venv/bin/activate` |
| Check | Your prompt now starts with `(.venv)`. From here on, plain `python` works on every system. | |

> **ACTIVATION IS PER TERMINAL WINDOW**
> Every time you sit down to work: open a terminal, `cd pinecrest-analysis`, activate, look for `(.venv)`. Forgetting is the number-one cause of `ModuleNotFoundError`.

## 3 Install the course packages

~10–20 min

Download **requirements.txt** from the course home page (Tools and Versions) into `pinecrest-analysis`. With `(.venv)` showing, run:

```
python -m pip install -r requirements.txt
```

It installs about twenty packages (NumPy, pandas, Matplotlib, seaborn, SciPy, statsmodels, JupyterLab, SQLAlchemy, Requests, Beautiful Soup, Selenium, Plotly, Bokeh, Streamlit and others) plus their dependencies. Success ends with a line starting **Successfully installed**.

## 4 The course data and the practice shop

~3 min

Download **pinecrest-data.zip** and **pinecrest-practice-site.zip** from the course home page (The Data) into `pinecrest-analysis`, then unzip both there. This works on every system:

```
python -m zipfile -e pinecrest-data.zip .
python -m zipfile -e pinecrest-practice-site.zip .
```

The `.` means "extract here". You now have a `data/` folder (twelve data files, a README and two programs) and a `practice/` folder, a copy of the practice web shop for Sessions 16 to 18 when the internet is unavailable. Pinecrest Outfitters is fictional: every customer, product and number is invented.

```
pinecrest-analysis/
├── .venv/              the environment (never edit by hand)
├── data/               orders.csv, customers.csv, pinecrest.db, ... and README.m
├── practice/           the offline practice shop
└── requirements.txt
```

In Session 1 your instructor hands out a `labs/` folder: it goes inside `pinecrest-analysis` too, next to `data/`.

## 5 JupyterLab or VS Code, and Chrome

~10 min

- **JupyterLab** (already installed): from `pinecrest-analysis` with the environment active, run `jupyter lab`. A browser tab opens; click **Python 3 (ipykernel)** to make a notebook. Stop it with **Ctrl+C** in the terminal.
- **Or VS Code**: install the **Python** and **Jupyter** extensions, **File ▸ Open Folder** on `pinecrest-analysis`, and pick the kernel and interpreter that mention `.venv`.
- **Google Chrome**: install it normally (google.com/chrome). Selenium finds it and fetches its driver by itself the first time, as long as you are online. If Plotly later says it cannot find Chrome to save an image, run `plotly_get_chrome` in the activated environment.

## 6 Run the environment check

~2 min

Your instructor sends `check_environment.py` with this guide. Save it in `pinecrest-analysis` (next to `data/`) and run it with the environment active:

```
python check_environment.py
```

It checks four things: Python is 3.12 or newer, an environment is active, every course package imports (with its version), and the data folder has all twelve files and 20,302 orders. Its last line is either **READY for Python Data Science** or **NOT READY** followed by a numbered list of exactly what to fix.

> ✅ YOU'RE READY
>
> Last line **READY for Python Data Science**? Reply to your instructor with that line, and you are set for Session 1. **NOT READY**? Try the fixes it lists and the table below, then send the whole output.

## COMMON INSTALL PROBLEMS

| YOU SEE… | TRY THIS |
|---|---|
| Windows: `python` is not recognized, or it opens the Microsoft Store | Before an environment is active, Windows only knows `py`. Create the environment with `py -m venv .venv`; once it is active, `python` works. |
| PowerShell: "running scripts is disabled on this system" | Run `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned` once, answer Y, and activate again. |
| Debian/Ubuntu: "ensurepip is not available" | `sudo apt install python3-venv python3-pip` (or `python3.12-venv`), delete the half-made `.venv` folder, create it again. |
| pip: "Could not find a version that satisfies the requirement…" | Check `python --version`: the course needs 3.12 or newer. Install it, delete `.venv`, then repeat Steps 2 and 3. If your Python is much *newer* than the course, remove the `~=…` part from the line pip names and install again. |
| `ModuleNotFoundError: No module named 'pandas'` | The environment is not active in this terminal (no `(.venv)`). In a notebook: the kernel is another Python; start `jupyter lab` from the activated terminal, or choose the `.venv` kernel in VS Code. |
| `FileNotFoundError` for `data/orders.csv`, or the check | Python is working in another folder, or the zip was unpacked into a nested folder. You need `pinecrest-analysis/data/orders.csv`; run |

| YOU SEE… | TRY THIS |
|---|---|
| says "data folder not found" | from `pinecrest-analysis`. |
| `jupyter lab` starts but no browser opens | Copy the `http://localhost:8888/lab?token=…` line from the terminal into your browser. |
| The install is slow or stops halfway | Run the same command again on a better connection: pip skips what it already has. |

Environment badly broken? Delete the `.venv` folder and repeat Steps 2 and 3. Your notebooks and data are not inside it, so nothing of yours is lost.

## THE COURSE AT A GLANCE

| SESSIONS | YOU'LL LEARN |
|---|---|
| 1–3 | Your toolkit, then NumPy: arrays, vectorized math, broadcasting, randomness, linear algebra |
| 4–8 | pandas: loading, cleaning, grouping, merging, reshaping, time series, windows, features |
| 8–10 | Matplotlib: line, scatter and bar charts, layouts, styling, saving figures |
| 10–13 | Statistics: summaries, distributions, correlation, hypothesis tests, an A/B test, an EDA routine (Checkpoint 1) |
| 14–16 | SQL from Python: joins, CTEs, window functions, indexes; a document database |
| 16–18 | Web data: HTTP, scraping static and dynamic pages, API keys and tokens, polite clients |
| 19–20 | seaborn, Plotly, Bokeh and a Streamlit dashboard; you choose your capstone question |
| 21 | Capstone studio and showcase: the Pinecrest Report |

21 sessions of 3 hours. Between sessions you finish each lesson's project, its exercises, quiz and journal entry on the course site: about two to three hours a week.

## FREQUENTLY ASKED

### CAN I USE ANACONDA, COLAB OR A SETUP I ALREADY HAVE?

Yes, if you end up with Python 3.12 or newer and the versions in `requirements.txt`, and the check says READY. Demos in class use a `.venv`, and a hosted notebook service also needs the data folder uploaded.

### DO I NEED ACCOUNTS, API KEYS OR PAID SOFTWARE?

No. Every tool is free, and the web sessions use the course's own practice shop and a practice API that runs on your laptop, with demo keys given in class.

### IS THE PINECREST DATA REAL?

No. It was generated for the course (`data/make_data.py` rebuilds it) to behave like real retail data, flaws included.

### WHAT IF A STEP FAILS AND I CANNOT FIX IT?

Come anyway, with the output of `check_environment.py`. Session 1 opens with an environment check and help.

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